

ORION

PART 08

Digital Twin & Simulation

Simulating infrastructure, disasters, network failures and scenarios

Technical Architecture Series • Beginner → Professional

How to use this volume

This document is one independent volume in the twelve-part ORION technical series. Beginners should read from the start; experienced readers can jump directly to the relevant architecture sections.

ORION is a secure communications, infrastructure-coordination, emergency-response, and decision-support architecture. High-impact weapons-control functionality is deliberately outside this design.

Purpose

Represent infrastructure and operational state in software so ORION can test strategies without manipulating real infrastructure.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Twin layers

Geography → topology → capacity → dependencies → incident state → simulation engine → AI planning/analysis.

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Represented state

Network nodes/links, compute, storage, connectivity constraints, service dependencies, incident states, and recovery policies.

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Scenarios

Network outage, regional connectivity loss, capacity spike, data-center failure, cloud dependency outage, and synthetic security scenarios.

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Scenario engine

A scenario contains initial state, events, constraints, assumptions, and evaluation metrics. Reproducibility is essential.

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AI + twin

Generate candidate strategies, simulate each, compare latency/availability/cost/recovery, then present the results to authorized operators.

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Validation

Compare simulations with controlled tests and historical observations. Record model error and assumptions.

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Uses

Capacity planning, resilience engineering, architecture review, training, incident rehearsal, research, and technology evaluation.

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Architecture Review Checklist

- Requirements and assumptions are explicit.
- Component ownership and boundaries are documented.
- Normal, degraded, and failure states are defined.
- Security and privacy are architectural requirements.
- Capacity and scaling assumptions are measurable.
- Logs, metrics, traces, and audit events cover critical flows.
- Dependencies have explicit contracts and failure behavior.
- Important decisions record alternatives and trade-offs.
- Testing covers expected behavior and controlled failures.
- Operational runbooks exist for important incidents.

Key Takeaways

Digital Twin & Simulation should be implemented as a bounded layer of ORION, with explicit interfaces to the rest of the platform.

The goal is not maximum complexity. The goal is measurable resilience, security, maintainability, interoperability, and operational usefulness.

Glossary

- API — Application Programming Interface used for software-to-software communication.
- Edge — Computing placed near users or physical systems.
- Event — A record that something happened.
- Failover — Switching to a redundant component after failure.
- Idempotency — Repeating an operation does not create an unintended additional effect.
- NTN — Non-terrestrial network, commonly involving satellite infrastructure.
- Policy — A rule that determines what is allowed under stated conditions.
- RPO — Recovery Point Objective; acceptable data-loss window.
- RTO — Recovery Time Objective; target recovery time.
- Zero trust — Continuous verification instead of implicit trust based on network location.